

1. Project Title

[SYNTHETIC TEST DATA] AI-Driven Gas Detection System

Background

Synthetic test data background information on mine safety and gas monitoring.

2. Problem Statement

Synthetic problem statement regarding hazardous methane accumulation in underground seams.

3. Research Objectives

Objective 1: Deploy wireless mesh sensor nodes.

Objective 2: Real-time telemetry monitoring.

4. Literature Review

Synthetic literature review analyzing existing wired gas detection systems.

5. Proposed Methodology

Methodology includes IoT spatial mesh node prototyping and lab validation.

6. Work Plan

24-month project timeline: Months 1-6 design, Months 7-18 lab trials, Months 19-24 field test.

7. Expected Outcomes

Expected deliverables: Autonomous warning system and reduced venting latency.

8. Estimated Budget

Total Estimated Budget: 4,500,000 INR.

9. Manpower

Principal Investigator, 2 Senior Research Fellows.

10. Equipment

Gas calibration chamber, wireless transceiver modules.